Parth Ganeriwala, Ph.D.

SOFTWARE ENGINEER II · RESEARCH SCHOLAR
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Summary

Ph.D. computer scientist researching formal verification and assurance of learning-enabled autonomous systems, with 30 accepted peer-reviewed publications spanning transfer learning, computer vision, LLM-based tooling, and formal methods. Currently a Software Engineer II at Avidyne, building DO-178C safety-critical avionics software in embedded C++ and AI-assisted certification tooling – pairing production engineering discipline with applied ML research.

Education

Florida Institute of Technology — *Melbourne, Florida*
PH.D. IN COMPUTER SCIENCE: GPA - 4.0/4.0 — *January 2023 – May 2026*

- Research Interests: Formal Methods, Artificial Intelligence, Machine Learning, Deep and Transfer Learning, LLMs and Automation
- *Dissertation:* **What is Common Knowledge Across Domains? Investigating Shared Representations in Transfer Learning** *(Advisor: Dr. Siddhartha Bhattacharyya)*
- Honors: **Outstanding Student of the Year** (Honors Convocation 2025); Inducted Member of **Phi Kappa Phi**

Florida Institute of Technology — *Melbourne, Florida*
M.S. IN COMPUTER SCIENCE: GPA - 4.0/4.0 — *May 2022 – December 2023*

- Core subjects: Artificial Intelligence, Database Systems, Formal Methods, Advanced Software Engineering, Speech Recognition.

Birla Institute of Technology and Science, Pilani — *Dubai, United Arab Emirates*
B.E. IN COMPUTER SCIENCE: GPA - 3.5/4.0 — *August 2018 – December 2021*

- Thesis: *ERF-CondLaneNet: an Ego Car Lane Detection Framework*

Skills

Programming	Java, C/C++, Python, Visual Basic, Bash, MATLAB, LaTeX
Avionics & Safety-Critical	DO-178C, DOORS, Embedded C++, Real-time Systems, HIL Simulation, Unit/Integration/E2E Testing, Test Automation
Data Analytics	Jupyter, pandas, Dask, Statsmodels, Seaborn, MySQL, MongoDB, MySQL Workbench, Neo4j, Elasticsearch
Machine Learning Libraries	scikit-learn, XGBoost, OpenCV, nltk, pandas, scipy
Deep Learning / LLMs	TensorFlow, PyTorch, Keras, CUDA, TensorRT, BERT, GPT, Hugging Face Transformers, Hugging Face Accelerate, PEFT/LoRA, LangChain, LlamaIndex, Claude Code
Cloud & MLOps	AWS (S3, SageMaker, EC2), Ray, Weights & Biases, MLflow
Formal Verification	NuSMV/nuXmv, Uppaal, AGREE, PRISM, TLA+ - Coq (Class Projects)
Robotics & Systems Engineering	Robot Operating System (ROS), AADL, SysML, Gazebo, RViz, Agile Software Development
Other Tools	Docker, Git, Perforce, GitHub Actions, NLP Modules, Android Studio
Project Tools	Jira, Notion, Slack, Teams

Experience

Software and Systems Engineering, Avidyne — *Melbourne, FL*
SOFTWARE ENGINEER II — *May 2026 – Present*

- Design and implement **embedded C++ software** for avionics flight display systems, spanning **operating system services**, **device drivers**, and **touchscreen-based user interfaces**.
- Verify embedded software correctness through **unit and integration testing** and maintain requirements-to-implementation traceability in **DOORS**, supporting **DO-178C** and **FAA** certification audits.
- Led testing for the **Vantage 12 major software release**, validating **60 tests across 2 hardware configurations**, cutting the release cycle by **67%** compared with the previous minor release.
- Author **certification tests** for the **IFD** product line for a major avionics OEM partner, covering ∼**100 existing tests** plus new tests for updated requirements.
- Write **certification tests** for an ongoing aircraft-manufacturer program.
- Co-create an internal **Claude Code toolkit** for **DO-178C** certification work, with reusable skills for requirements lookup, test review, traceability audits, and test-log grading, plus **Linux** installer support; adopted by **5 engineers**.
- Built a **headless command-line mode** for the in-house **E2TE** C++/MFC test tool and an AI workflow that grades result logs and retries failed runs, targeting the ∼**20 minutes** of manual review per run.

- Traced root causes of **verification test failures** across test scripts, the test library, flight code, and emulator configuration using **Claude-assisted analysis**.
- Delivered an internal **configuration-generation tool** for Vantage 12 in **1 day** against a **2-week** assignment, now adopted by manufacturing.
- Mentor **6 early-career engineers**, including **4 interns**, guiding onboarding and technical growth.

Software and Systems Engineering, Avidyne

Melbourne, FL

SOFTWARE ENGINEER INTERN / CO-OP

May 2025 – May 2026

- Expanded **end-to-end (E2E) test and evaluation (E2TE)** coverage across navigation, communication, flight management, and flight display systems, increasing overall test coverage by **25%**.
- Developed, integrated, and debugged **C-based flight software** in the avionics stack, supporting compliance with real-time, safety-critical **DO-178C** guidelines for production-ready systems.
- Built **30+ system-level and flight-specific test cases** and **TFC (Test For Credit)** software tests, reducing initial verification cycle time by **10%**.
- Consolidated product test libraries across multiple avionics platforms into a unified cross-product testing infrastructure, improving code reusability and maintenance efficiency by ∼**30%**.
- Automated regression and validation testing by writing **C and Visual Basic** test scripts (managed via **Perforce** and **Visual Studio**), cutting simulation turnaround time by up to **20%**.
- Diagnosed and resolved execution issues in **hardware-in-the-loop (HIL) simulations** using proprietary avionics simulation frameworks, improving simulation-to-aircraft fidelity by ∼**10%**.
- Supported flight code development for Avidyne's **Quantum Open Avionics Platform**, supporting rapid prototyping of next-generation avionics solutions.

ASSIST Research Lab, Florida Institute of Technology

Melbourne, FL

RESEARCH SCHOLAR

August 2021 - Present

- Working with a team of research professionals for formal methods of verification and run-time assurance, ML, IoT, robotics, and cyber security.
- Helped develop **research proposals with professors** on diverse topics, including cognitive agents on human behavior, the assurance of artificial intelligence in safety-critical systems, and the fine-tuning of LLMs for domain-specific queries.
- Collaborating on NASA's ***University Leadership Initiative (Round 8)*** as part of a **Florida Tech**-led multi-university/industry team to develop a framework for **trustworthy, increasingly autonomous aviation safety systems**; partners include ***Penn State, NC A&T State, UF, Stanford, Santa Fe College, Uni of New Mexico, Collins Aerospace, and ResilienX***; part of awards totaling up to $6.9M over three years.
- Modeling human cognitive behavior representation formally with respect to cyber-sickness in AR/VR systems, with ***Collins Aerospace, Iowa State University, RTX Technologies Research Center (RTRC), and Smart Information Flow Technologies (SIFT)***, funded by ***DARPA***.
- Applied **Large Language Model (LLM) translation** to cognitive architectures with ***Penn State University***, focusing on enhancing the integration of LLMs to facilitate communication and knowledge transfer within cognitive systems.
- Developing a modeling approach for designing, maintaining, and supporting air and sea platform fiber optic communications technology with ***Critical Frequency Design***, funded by ***Naval Air Systems Command***.
- Formally verifying the autonomous agent to assure safety as well as the logical correctness of the safety-critical system, with ***Rockwell Collins and Soar Tech***, funded by ***NASA***.
- Advising a Ph.D. student and undergraduate students on transfer learning, automated data labeling, and assurance frameworks for vision-based classification in autonomous aircraft systems funded by ***NASA***, addressing safety and reliability challenges in aviation technologies.
- Mentoring **20–30 undergraduate and high school students** on machine learning engineering approaches in the aerospace and systems domains, leading to **11 publications** that addressed real-world challenges in these fields.
- Investigated the development of a **cognitive-enhanced agent** for automatically piloting aircraft in dense urban environments which emphasized safe and reliable takeoff/landing among aerial traffic without human intervention.
- Created **AssistTaxi**, a novel dataset for runway and taxiway analysis published at IEEE ICMLA 2023 and extended as AssistTaxi-v2 (AIAA SciTech 2026), contributing to autonomous operations.
- Developed an **autonomous aircraft perception system** for accurately detecting and labeling line markings on an airport taxiway.

L3Harris Institute for Assured Information, Florida Institute of Technology

Melbourne, FL

GRADUATE RESEARCH ASSISTANT

May 2024 – June 2024

- Developed a **decentralized framework** enabling **multiple autonomous agents**—robotic dogs, drones, and mobile robots—to coordinate, communicate, and reach shared goals.
- Collaborated with **developers and professors** to rigorously test the system in both **simulation** and **real-world environments**.

IRI Research, Florida Institute of Technology

Melbourne, FL

IRI RESEARCH DEVELOPER

May 2023 - July 2023

- Proposed and **implemented a framework using AI language models** to automatically extract software requirements from source code.
- Supervised and coordinated with undergraduate students towards the development process.

Publications

Why Trust Machine Learning? Evolution of Trustworthiness Debt Leveraging Trustworthiness Attributes

Under Review

KMAL FARABI, A PRITCHETT, M SYNBORSKI, J LIU, M CARROLL, M NAMUKASA, N MANNING, R SEN, MAH KHAN, **P GANERIWALA**, S BHATTACHARYYA, S RAY, S BHUNIA, M MAHMOUD, A HOMAIFAR, P KEBRIA, M PAVONE, M GANAI AND C LIU

AAAI 2027

It Is Not the Model, It Is the Input: Test-Time Input-Space Refinement for Few-Shot Segmentation

Under Review

MAH KHAN, **P GANERIWALA**, A ALVAREZ AND S BHATTACHARYYA

AAAI 2027

Surveying the Landscape of Transfer Learning: Common Knowledge and Beyond

P GANERIWALA, N NEOGI AND S BHATTACHARYYA

Under Review

IEEE Trans on Pattern Analysis and Machine Intelligence

Formalizing Natural Language Specifications for MBSE with an NLP-Based Grammar for Automated System Architecture Generation

C CHAMBERS, **P GANERIWALA**, S MUELLER, S BHATTACHARYYA AND C SEN

Under Review

IEEE Systems Journal

VIROPLEX: A GNN-Driven Assessment of the Pandemic Potential of Zoonotic Bat Viruses

A AGNIHOTHRAM, **P GANERIWALA**, AND S BHATTACHARYYA

Accepted

IEEE ICTAI 2026

Formal Analysis of Stochastic Cognitive Models: A Translation Framework from Soar to PRISM

P GANERIWALA, C CHAMBERS, JI LATHROP, S BHATTACHARYYA, J BABAR, S GILBERT, I AMUNDSON, M DORNEICH AND MAH KHAN

Accepted

NASA Formal Methods 2026

Compositional Reasoning over System Architectures with Integrated Cognitive Model

P GANERIWALA, C CHAMBERS, S BHATTACHARYYA, I AMUNDSON, AND J BABAR

Accepted

IEEE SysCon 2026

Correlation-Threshold Feature Selection for Internet of Things (IoT) Device Identification under Domain Shift

A DHANAWADE, **P GANERIWALA**, AND S BHATTACHARYYA

Accepted

ISAIM 2026

AssistTaxi-v2: A Scalable Dataset for Taxiway/Runway Scene Understanding Under Diverse Conditions

P GANERIWALA, MAH KHAN, A ALVAREZ, S BHATTACHARYYA, N NEOGI AND S LEHMAN

Accepted

AIAA SciTech 2026

Explainable Assurance through Compositional Verification with Cognitive Models

P GANERIWALA, C CHAMBERS, S BHATTACHARYYA, AND J BABAR

Accepted

IEEE RTSS/ESRA 2025

Evaluating LLM Translation for Prompt-Enhanced ACT-R and Soar Models

P GANERIWALA, S WU, S BHATTACHARYYA AND F RITTER

Accepted

BRIMS 2025

Enabling Formal Verification in a Common Model of Cognition

P GANERIWALA, M MATSUMURO, F RITTER AND S BHATTACHARYYA

Accepted

BRIMS 2025

Integrating Reconfigurable Accelerators with Quantum Computing

PRATIBHA, **P GANERIWALA** AND N MAHMUD

Accepted

IEEE QCE QCore Workshop 2025

Adapt, But Don't Forget: Fine-Tuning and Contrastive Routing for Lane Detection under Distribution Shift

MAH KHAN, **P GANERIWALA**, S LEHMAN, A ALVAREZ, S BHATTACHARYYA AND N NEOGI

Accepted

ICCV 2COOOL Workshop 2025

AI Driven Differentiation and Quantification of Metal Ions Using ITIES Electrochemical Sensors

M AHMED, **P GANERIWALA**, A SAVVIDOU, N BREEN, S BHATTACHARYYA, P PATHIRATHNA

Accepted

Journal of Sensor and Actuator Networks 2025

FLAIR: Few-Shot Learning for Grapheme Recognition in Ancient Scripts

P GANERIWALA AND D MITRA

Accepted

CVPR SINT4CH Workshop 2025

Few-Shot Learning for Grapheme Recognition in Ancient Scripts

P GANERIWALA AND D MITRA

Accepted

ACM Journal on Computing and Cultural Heritage 2025

Can Someone Prove Your Operator Won't Get Distracted? A Gentle Introduction to Formal Methods in Human Factors

S GILBERT, **P GANERIWALA**, J LATHROP, A NEWENDORP, S FIEFFER, P WU, I AMUNDSON, C CHAMBERS, A KOHL, S KHAN, M SANAEI, J BABAR, T WANG, D MUSLINER, R GOLDMAN, J GOTTLIEB, E WINER, M DORNEICH AND S BHATTACHARYYA

Accepted

HFES 2025

Modeling and Formal Analysis of High-Assurance Mixed-Reality Systems

I AMUNDSON, J BABAR, H HERENCIA-ZAPANA, S F ROLLINI, B BRUSSEE, P WU, T WANG, D MUSLINER, R GOLDMAN, J GOTTLIEB, A NEWENDORP, A KOHL, S FIEFFER, S KHAN, M SANAEI, M MUSCALA, S GILBERT, E WINER, M DORNEICH, J LATHROP, **P GANERIWALA**, C CHAMBERS AND S BHATTACHARYYA

Accepted

AIAA DATC/IEEE DASC 2025

Systems Engineering with Architecture Modeling, Formal Verification and Human Interactions for Learning-Enabled Autonomous Agent

P GANERIWALA, R JONES, M MATESSA, S BHATTACHARYYA, J DAVIS, S ROLLINI, H PUROHIT, N NEOGI

Accepted

INCOSE Systems Journal

Design and Validation of Learning Aware HMI for Learning-Enabled Increasingly Autonomous Systems — *Accepted*

P GANERIWALA, M MATESSA, S BHATTACHARYYA, R JONES, J DAVIS, P KAUR, S ROLLINI, N NEOGI — *IEEE SysCon 2025*

Automating Physics-Based Reasoning for SysML Model Validation — *Accepted*

C CHAMBERS, S MUELLER, **P GANERIWALA**, S BHATTACHARYYA AND C SEN — *IEEE SysCon 2025*

Runway vs. Taxiway: Challenges in Automated Line Identification and Notation Approaches — *Accepted*

P GANERIWALA, A ALVAREZ, A ALQAHTANI, S BHATTACHARYYA, MAH KHAN, N NEOGI — *IEEE SysCon 2025*

Exploring Machine Learning Engineering for Object Detection and Tracking by Unmanned Aerial Vehicle (UAV) — *Accepted*

A GUNA, **P GANERIWALA**, AND S BHATTACHARYYA — *IEEE ICMLA 2024*

ALINA: Advanced Line Identification and Notation Algorithm — *Accepted*

MAH KHAN, **P GANERIWALA**, S BHATTACHARYYA, N NEOGI AND R MUTHALAGU — *CVPR VDU Workshop 2024*

AssistTaxi: A Comprehensive Dataset for Taxiway Analysis and Autonomous Operations — *Accepted*

P GANERIWALA, S BHATTACHARYYA, S GUNTHER, B KISH, MAH KHAN, A DHADOTI AND N NEOGI — *IEEE ICMLA 2023*

Towards Knowledge Extraction and Parsing of XML Metadata for SysML System Architecture Modeling — *Accepted*

C CHAMBERS, **P GANERIWALA**, S BHATTACHARYYA, C SEN AND N NUR — *IEEE UEMCON 2023*

Automated Framework to Extract Software Requirements from Source Code — *Accepted*

C MISKELL, R DIAZ, **P GANERIWALA**, K SLHOUB, F NEMBHARD — *ACM NLPIR 2023*

Assuring Learning-Enabled Increasingly Autonomous Systems (ALEIAS) — *Accepted*

N NARAYAN, **P GANERIWALA**, R JONES, M MATESSA, S BHATTACHARYYA, J DAVIS, H PUROHIT AND S ROLLINI — *IEEE SysCon 2023*

IPAssess: A Protocol-Based Fingerprinting Model for Device Identification in IoT — *Accepted*

P GANERIWALA, S NANDANWAR, A GUPTA, S BHATTACHARYYA AND R MUTHALAGU — *SAI IntelliSys 2023*

Cross Dataset Analysis and Network Architecture Repair for Autonomous Car Lane Detection — *Accepted*

P GANERIWALA, S BHATTACHARYYA AND R MUTHALAGU — *IEEE Intelligent Vehicles Symposium (IV) 2023*

Functional Reasoning of System Architecture in the System Modeling Language (SysML) With XML Representation — *Accepted*

P GANERIWALA, C CHAMBERS, C SEN AND S BHATTACHARYYA — *ASME IDETC-CIE 2023*

Modeling IoT Behavior for Enforcing Security and Privacy Policies — *Accepted*

A GUPTA, D CAMPOS, A DCOSTA, **P GANERIWALA**, S BHATTACHARYYA AND T OCONNOR — *SAI Computing Conference 2022*

Toward Generating System Architecture and Formal Functional Description in the Architecture Analysis & Design Language (AADL) With Structured Natural Language — *Accepted*

A CHAUHAN, **P GANERIWALA**, C SEN AND S BHATTACHARYYA — *ASME IDETC-CIE 2022*